

Dr. Jagrati Talreja, PhD

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SUMMARY

AI Research Scientist and Machine Learning Engineer with 5+ years of experience in computer vision, deep learning, multimodal AI, and data analytics. Expertise in building CNN, Transformer, GAN, and diffusion-based systems for image enhancement, remote sensing, SAR-optical translation, and satellite/UAV data fusion. Strong background in Python, PyTorch, GPU training, experimental design, and research-to-production style prototyping. Published in IEEE Transactions on Geoscience & Remote Sensing and experienced in federally funded NASA/NSF research environments.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS, Node.js, Bootstrap, MATLAB.

ML/AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, Keras, CNNs, Vision Transformers, GANs, Diffusion Models, Autoencoders, PINNs, Transfer Learning, XAI, Segmentation, Classification, Regression, Super-Resolution.

Data & Geospatial: Remote Sensing, Sentinel-1, Sentinel-2, UAV imagery, multimodal fusion, image tiling, geospatial preprocessing, Google Earth Engine, Alaska Satellite Facility, ArcGIS Pro, SNAP.

Infra & Tools: CUDA, cuDNN, Jupyter, VS Code, Linux, WSL, Git, Overleaf, Office 365, PyCharm, Anaconda.

Hardware/Systems: NVIDIA A4000/A6000, A100/H100 clusters, FPGA, Raspberry Pi, SoC systems, Arduino.

EXPERIENCE

Postdoctoral Fellow / Co-Lead / Co-PI, NASA & NSF Funded Projects <i>North Carolina A&T State University</i>	Jan 2025 – Present <i>Greensboro, NC, USA</i>
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- Co-I on NASA COMPASS proposal (2026) for AI-driven remote sensing.
- Led end-to-end development of multimodal deep learning pipelines for flood mapping across SAR and optical datasets, improving model performance and scalability on GPU clusters.
- Designed and deployed CNN/Transformer/GAN pipelines improving PSNR by +2.2 dB (40% reduction in reconstruction error) on 20,000+ multimodal satellite image tiles using PyTorch on multi-GPU clusters.
- Trained models on A4000/A6000 GPU workstations for large-scale satellite data.
- Engineered scalable geospatial ML pipelines.
- Collaborated with NASA-affiliated teams; mentor students.
- Developing quantum-inspired ML pipelines, exploring spatial feature encoding for next-generation ML systems.

PhD Researcher, Electrical Engineering <i>Chulalongkorn University</i>	Aug 2020 – 2024 <i>Bangkok, Thailand</i>
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- **FPGA Super-Resolution:** Real-time DL model for FPGA-based SoC deployment.
- **Signature Verification:** Shape-based authentication system.
- **Embedded Vision:** OpenCV-based tracking on Raspberry Pi.
- Mentored CV/DSP labs (Teaching Assistant).

Research Assistant / Intern <i>Indian Institute of Technology</i>	Jun 2018 – Jun 2019 <i>Kanpur, India</i>
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- Built MATLAB-based motion detection system named Disruption Detector.
- Applied computer vision for image acquisition and event detection.

Engineering Projects (Undergrad) <i>Pranveer Singh Institute of Technology</i>	Aug 2015 – Dec 2018 <i>Uttar Pradesh, India</i>
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- **Medi-Cart:** RFID-based healthcare system for patient tracking.
- **Robo Car:** Arduino-based voice-controlled robotic system.
- **Aircraft Prototype:** Transmitter-based aircraft development.
- **Android Apps:** Mobile app development using Android Studio.
- **Software System:** Java + MySQL-based academic management platform.

EDUCATION

PhD in Data Science & Artificial Intelligence <i>Chulalongkorn University</i>	2019 – 2024 <i>Bangkok, Thailand</i>
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Thesis: *Single Image Super-Resolution using Deep Attention Networks.*

- C2F Scholarship (PhD, 2019), Chulalongkorn University.
- Overseas Research Experience Scholarship, 2024.
- 90th Anniversary Scholarship, Chulalongkorn University.

B.Tech in Electronics & Communication Engineering <i>Dr. A.P.J. Abdul Kalam Technical University</i>	2015 – 2019 <i>Lucknow, India</i>
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- Gold Medal for excellence in Institutional Academia with 80.93%.
- Final-year project resulted in an Indian patent.
- Selected among the top 100 Engineering & Science projects in Uttar Pradesh (2019).
- Awarded University and State Government Scholarships for 4 years.

ACHIEVEMENTS

- GRE: 304 (2020).
- IELTS Academic: 6.5 Overall (2019).
- IIT JEE Mains Qualified (2015).

CONFERENCES, SPEAKING & LEADERSHIP

- Speaker, International Geoscience and Remote Sensing Symposium (IGARSS), August 10-15, 2026.
- Speaker, American Society for Photogrammetry and Remote Sensing (ASPRS), February 15-19, 2026.
- Speaker, American Geophysical Union (AGU), December 15-19, 2025.
- Speaker, GIS Day, November 19, 2025.
- Host, IEEE R10 Robotics Competition for Managing Climate Change, Chulalongkorn University (2023).

PATENTS

- **SafePath Navigate to Safety**, US Application No. 63/929,945, 2025.
- **Medi-Cart: Telemetry using RFID**, Indian Patent No. 201811037266A, 2018.

PUBLICATIONS

- **J. Talreja**, T.S. Gebre, L.H. Beni, “FLoRA: Fusion-Latent for Optical Reconstruction and Flood Area Segmentation via Cross-Modal Multi-Task Distillation Network,” *IEEE Transactions on Geoscience & Remote Sensing*, 2026.
- **J. Talreja**, T.S. Gebre, L.H. Beni, “DPolTransGAN: Scattering Guided Hybrid Transformer-GAN for SAR-Optical Image Translation using Dual Polarization SAR,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2026 .
- **J. Talreja**, T.S. Gebre, L.H. Beni, “Cross-Polarization Fusion of VV and VH SAR Observations for Improved Flood Mapping,” *IGARSS*, 2026.
- **J. Talreja**, S. Aramvith, T. Onoye, “DANS: Deep Attention Network for Single-Image Super-Resolution,” *IEEE Access*, 2023.
- **J. Talreja**, S. Aramvith, T. Onoye, “DHTCUN: Deep Hybrid Transformer Network for Single-Image Super-Resolution,” *IEEE Access*, 2024.
- **J. Talreja**, S. Aramvith, T. Onoye, “XTNSR: Xception-Based Transformer Network for Single-Image Super-Resolution,” *Complex & Intelligent Systems*, 2024.
- **J. Talreja**, L.H. Beni, “SARGANet: A GAN-U-Net Hybrid Framework for Super-Resolution of Sentinel-1 SAR Imagery,” *Earth Science Informatics*, 2026.
- **J. Talreja**, T.S. Gebre, L.H. Beni, “Region8: Spatially-Aware Quantum Encoding for Flood Patch Classification in Multi-Modal Remote Sensing,” Under Review.
- **J. Talreja**, L.H. Beni, “STCN-SR: A Spatial Transformer-CNN Framework for Enhancing Sentinel-2 Imagery,” Under Review.
- T.S. Gebre, **J. Talreja**, L.H. Beni, “Overcoming" Physics Shock" in Earth Observation A Heteroscedastic Uncertainty Framework for PINN-based Flood Inference,” Under Review.